

Biometric Methods for User Research: Three Case Studies

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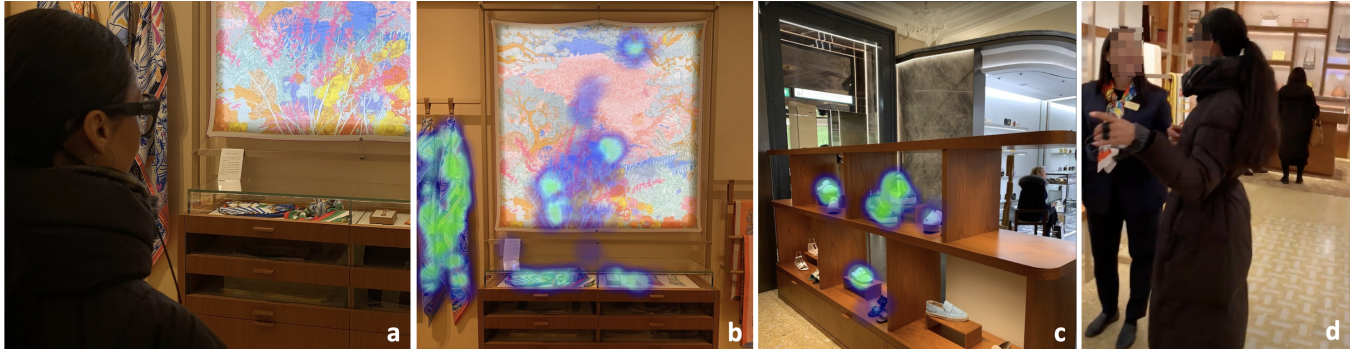


Figure 1: Case Study 3: (a) Participant looking at the foulards; (b) Eye-tracking heatmaps at the foulards; (c) Eye-tracking heatmaps at the product display shelf; (d) Participant consulting a shop assistant.

ABSTRACT

Understanding user experience (UX) and its impact on user preferences and purchasing decisions, and loyalty is essential in today's user-centric business landscape. Assessing UX during product trials is vital for the design and improvement of products. While traditional UX evaluation methods (e.g., questionnaires and observations) offer valuable insights into user behavior, they rely on indirect, subjective and non-real-time data, making them prone to bias. There is a growing interest in incorporating neuroscience techniques into UX research to measure users' neurophysiological responses to design elements and interactive experiences. We evaluated two digital prototypes and one in-store experience in three small-sample-size biometric UX studies that can fit into time-constrained product development processes. Our findings demonstrate the potential of biometric methods to complement existing UX research approaches. We discuss the lessons learned and advocate for the establishment of research ethics committees within companies and standardization strategies for conducting biometric user research.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Usability testing; Field studies.**

KEYWORDS

Biometric User Research, Biometric Sensors, EEG, GSR, Eye-tracking

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1 INTRODUCTION

Understanding user experience (UX) and its impact on user preferences, purchasing decisions, and loyalty is vital in today's competitive and user-centric business landscape. Assessing UX during product trials holds practical significance for product design and improvements. It serves as the foundation for successful product development, enabling companies to align their offerings with what users truly want, ensuring product intuitiveness and user-friendliness [22].

Traditional evaluation methods for assessing user experiences and product usability usually rely on inquiries, such as questionnaires or think-aloud protocols [10], field methods like behavioral observation and ethnographic field studies [9], or performance metrics such as task completion time and number of errors [21]. While these methods offer valuable insights into user behavior, they also have limitations. Inquiries and field methods rely on indirect and subjective data, which may be affected by scale ambiguities and are difficult to gain real-time insights without interrupting the interaction [23, 31]. Moreover, performance metrics are mostly quantitative. They do not provide much information about users' mental states and emotional experiences [3, 21]. Moreover, these evaluation methods are based mainly on the explicit and sometimes biased responses of users [23] and cannot measure users'

subconscious processes, which, in contrast, can become evident when examining physiological and neuroimaging data [25].

There has been a growing interest in incorporating neuroscience techniques into Human-Computer Interaction (HCI) to measure human neuro-physiological responses to design elements and interactive experiences. These techniques offer complementary information to traditional evaluation methods and have the potential to address the aforementioned limitations [18]. Neuroscience techniques typically make use of biometric sensors, including electroencephalography (EEG), eye-tracking, facial muscle activity determined by Electromyography (EMG), and electrodermal activity (EDA), also known as Galvanic Skin Response (GSR) [3]. These sensors enable researchers to gain insights into users' cognitive, emotional, and physiological reactions when interacting with (digital) products or environments.

2 BIOMETRIC SENSORS

2.1 Eye-tracking, EEG, and EDA

Eye tracking is a method used to measure and analyze patterns of visual attention. It enables real-time monitoring of individuals' gaze points, the duration of their fixations on specific visual stimuli, and the trajectory of their eye movements. This method has been applied in various research fields, including human factors, cognitive psychology, marketing, and HCI. It serves as a valuable tool for investigating and understanding the intricate mechanisms underlying visual attention and information processing [4]. Companies have been adopting small-sample-size qualitative eye-tracking studies that can fit into tight development timelines and highly focus on specific interface components, while simultaneously supporting larger-scale eye-tracking studies to gain a holistic view of the entire product experience [4].

EEG, or electroencephalography, measures the electrical activity in the brain. When neurons in the brain are active, they emit a small electric current. Using sensors placed on the human scalp, EEG can pick up these signals and provide insights into various aspects of brain function, including brainwaves categorized into frequency bands (e.g., delta, theta, alpha, beta, and gamma) associated with different cognitive states. For example, alpha waves indicate a relaxed but awake state, while delta waves are observed during deep sleep [3]. Since EEG can detect changes in brain activity milliseconds after they occur, it enables researchers to interpret associated user experiences in real-time [29]. However, it remains difficult to pinpoint the exact source of specific brain activity due to the limited spatial resolution of EEG. Still, it is possible to identify the general source of EEG signals, such as left-sided, right-sided, and central activation in occipital, parietal, temporal, central, and frontal areas [14].

EDA stands for Electrodermal Activity, measures the electrical conductance of the skin, often referred to as the skin's "sweat response" or "galvanic skin response (GSR)". Measurements of EDA include both skin conductance level (SCL) and rapid phasic components (SCRs), which can be indicators of physiological or psychological arousal [5]. In simpler terms, it records how much a person is sweating, which can vary based on emotional states such as stress, excitement, or anxiety [19]. In a commercial context, measuring EDA is valuable for assessing consumers' emotional reactions and

predicting their decision-making processes based on their level of arousal. Importantly, measurements of EDA can reveal these responses even when people are not consciously aware of them [3].

2.2 Eye-tracking, EEG, and EDA in User Research

There has been a growing trend in consumer and HCI research, either adopting a single physiological sensor or combining multiple sensors to understand users' perceptions, preferences, emotional responses, and experiences with products. In a recent study, Wang et al. [32] explored the correlation between eye-tracking data and product aesthetic ratings, discovering that four eye-tracking indices (total fixation duration, fixation count, visit count, and total visit duration) significantly correlated with product aesthetic ratings. In their review paper, Wedel and Pieters [33] emphasize how eye-tracking offers unique insights into communication processing, challenges existing beliefs about the visual communication performance, and serves as a valuable methodology for tracking visual marketing effectiveness. On more specific product attributes, Du and McDonald [8] tests whether eye-tracking data can predict the relative importance of features, with respect to their visual design (e.g., visual appearance and size). The results demonstrate a significant correlation between feature importance and various gaze data.

In addition to eye-tracking techniques, Songsamoe et al. [29] applied EEG in evaluating consumer preferences for food associated hedonic evaluation. They analysed the asymmetry of EEG power spectrum and found that changes in alpha and beta bands reflected shifts in emotional states and brain activity correlating with food acceptability. Semenova et al. [27] found that both perceived taste and EEG beta power at the frontal region of the brain (indicating heightened cognitive engagement) significantly and independently correlated with "willingness to pay" for chocolates.

Besides food experiences, Kalantari et al.'s study [17] examined the influence of retail store layouts on consumer experiences, integrating subjective feedback and EEG data. They found that, despite no significant differences in subjective evaluations of layout complexity, more intricate layouts resulted in increased cognitive workload, highlighting the value of EEG data in complementing our understanding of consumers. Moreover, Abujelala et al. [1] introduced the Brain-EE system for evaluating user-engaged enjoyment, utilising a consumer-grade EEG tool (Muse). They found that an increase in theta activity in the frontal brain region strongly and reliably predicted the preferred game for each user.

Ayata et al. [2] proposed a music recommendation framework that classifies users' emotions using data from galvanic skin response (GSR) and photoplethysmography (PPG) sensors, demonstrating high accuracy in emotion-based music recommendations. Hernandez et al. [13] leveraged an EDA sensor to recognize ease of engagement of children during a naturalistic social interaction with an adult. Neto et al. [24] observed that EDA can serve as a measure of emotional arousal and can be translated into consumer engagement with cosmetics product advertisements.

The aforementioned studies mainly adopted one type of biometric sensor. Researchers have proposed combining techniques to triangulate the collected data for useful predictions [3, 35]. Guo et

al. [12] integrate eye-tracking and EEG measurements to examine the visual aesthetics of a product. Steinert et al. [30] present an exploratory study that triangulated data from EEG and electrocardiogram (ECG) and modeled the relationships between design behavior, problem-solving preferences, and real-time physiological data of engineers. Zhu and Lv [35] point out the challenges of adopting biometric sensors in user research, such as lack of a user research system and limitations in terms of experimental feasibility in outdoor environments, long preparation time of EEG equipment, and errors in physiological signal acquisition. They recommend that biometric UX research should integrate various measurements, including EEG, eye-tracking, skin electrical, respiration, and heart rate signals.

We will present three case studies demonstrating the experimental setup and data collection using a combination of biometric sensors to uncover user experiences in three commercial contexts: desktop analytical dashboard evaluation, automotive mobile application evaluation, and in-store evaluation outside the lab environment. These case studies are based on a small sample size of maximum 8 subjects, following a qualitative research approach [4]. The decision to utilise a qualitative small-sample-size research methods was intentional, which is driven by time constraints and allows us to delve into not just “what” but “why” and “how” individuals interact with the interfaces and environments. To support product development, our approach emphasizes rich, context-specific insights over generalizability.

3 CASE STUDY 1: DESKTOP ANALYTICAL DASHBOARD EVALUATION WITH BIOMETRIC SENSORS

3.1 Project Context

Our client in the insights and consultancy industry aimed to enhance their analytical dashboard’s user experience with a new layout. This dashboard is vital for brand tracking, presenting current brand trends and predicting future ones. To evaluate the new design, we conducted usability testing using biometric sensors (i.e., EEG, eye-tracking, GSR, and webcam facial expression analysis). This case study aimed to compare the old and new versions of the dashboard by measuring users’ cognitive workload, engagement, attention, confusion, and stress levels during interactions.

3.2 Methods

3.2.1 Participants. We recruited eight participants (7 females, mean age 27.5 ± 6.5). Two participants had seen the dashboards but hadn’t used them. All of them underwent a pre-screening process for eligibility, ensuring no neurological or psychiatric disorders, and had normal or corrected vision. They participated voluntarily without compensation.

3.2.2 Experimental procedure. The in-person study was moderated and lasting approximately 60 minutes. Participants were welcomed into the testing room, seated comfortably in front of a desktop screen connected to a testing laptop, and provided with

a mouse and keyboard as input devices. The experiment was conducted using iMotions software¹ and had six tasks that consist of short actions, which are helpful for data segmentation during the analysis phase. These tasks were: (1) observe the old dashboard without interaction, (2) observe the new dashboard without interaction, (3) create a new event and view the changes in the charts, (4) change country, market & category on the dashboard, (5) point out the changes made in Task 4, and (6) save the changes as a new view. The experimental procedure was as follows: **Step 1 (10 mins): Introduction and Informed consent.** Participants received a consent form outlining the study objectives and biometric sensors. They were instructed to keep their non-dominant hand still (where the GSR device was placed) to reduce signal noise, and carefully read and signed the consent form. **Step 2 (10 mins): Biometric sensors setup.** The experimenter set up the EEG cap and GSR device while ensuring participant comfort. **Step 3 (15 minutes): Eye-tracking calibration and baselines.** We performed eye-tracking calibration by having participants track a moving dot on the screen. Then, a 90-second relaxing video was shown to establish GSR baseline levels, and a resting-state EEG baseline was recorded as participants focused on a circle on the screen for 60 seconds. **Step 4 (25 mins): Performing six tasks and recording data.** The task instructions were both displayed on the screen and read aloud by the experimenter. Participants had sufficient time to complete each task.

3.2.3 Biometric sensors and montage. Three biometric sensors were employed in this study, along with a webcam and screen recording to collect the data (see Fig. 2a). iMotions facilitated the synchronization of data from these sensors: **(1) EEG device:** We used an Enobio 8 device (Neuroelectrics®) with 8 channels and a 500Hz acquisition frequency. This device was attached to a Neuroelectrics Neoprene Headcap on the participant’s head using a strap band. Eight NG Geltrode electrodes were positioned following the 10-20 Standard at F7, F8, T7, T8, Cz, O1, O2, and EOG1 locations. The reference electrode was on the left earlobe. A Necbox (hosting the EEG amplifier, battery and communications systems) was connected to a 10 Electrode Cable (8 electrodes plus two earlobe reference electrodes). We applied Signal Gel to each Geltrode and assessed data quality with NIC 2.1.3.3 software for Windows, ensuring either good (green) or average (orange) quality for each electrode. **(2) GSR device:** A Shimmer3 GSR + Unit was used to capture GSR data and estimate heart rate. It was placed on the participant’s non-dominant hand, with two GSR electrodes on the index and middle fingers and the pulse signal electrode on the ring finger. Data acquisition quality was verified through iMotions. **(3) Eye-tracking, webcam and screen recording:** A Tobii Pro Nano device was positioned at the bottom of the screen. A Logitech Webcam was placed on the screen to record participants’ facial expressions. Screen recording was conducted through iMotions and combined with eye-tracking data to analyze participants’ gaze patterns.

¹iMotion: The software allow researchers to collect and analyze human behavior data through eye tracking, galvanic skin response, facial expression analysis, EEG, EMG, and ECG on one convenient platform, <https://imotions.com>, accessed on 10 October 2023

3.2.4 Data preprocessing and analysis. We used iMotions software for experiment design and data collection from multiple biometric sensors. Each participant had seven recordings: one EEG baseline and six tasks. Before analysis, we visually inspected each recording with iMotions and segmented tasks. This granularity enabled detailed analysis of biometric metrics for shorter actions instead of averaging across the entire recording.

We used Python and iMotions for data preprocessing and analysis. GSR data underwent calculation of metrics (peaks per minute and number of peaks) in iMotions. Eye-tracking data were visually inspected using iMotions to study gaze behaviors and generate heatmaps, scan paths, and metrics like time-to-first-fixation and revisits. Facial expression analysis used the Affectiva AFFDEX algorithm [20] in iMotions to compute emotion percentages for each segment.

The EEG data, exported from iMotions in .csv format, were converted to FIF format and preprocessed using MNE-Python. The preprocessing steps included high-pass filtering (1Hz), low-pass filtering (100Hz), and a 50Hz notch filter to remove line noise. Then, the data were segmented into 2-second epochs and a global rejection threshold was calculated using the library Autoreject [16]. Next, the data were visually inspected to remove the moving artifacts in epochs, and an independent component analysis (ICA) was applied (infomax algorithm, 7 components, random state = 99). The computed ICA components were then inspected in their spatial and temporal domains, and components related to blinks or eye movements were identified. ICA was applied to the signal to regress out eye movements and blinks. The cleaned EEG data was used to calculate cognitive load [11] as a proxy measure of task engagement. We first computed the power spectral density (PSD) of the EEG data (Welch method, 1-second long segments, 50% overlapping window). The PSDs were computed for each epoch, in 5 frequency bands (delta, theta, alpha, beta, and gamma). Then, the computed PSDs were averaged across epochs, resulting in each subject having 5 PSD values per electrode and segment. Finally, cognitive load was computed as the averaged PSDs in the theta band between frontal electrodes F7 and F8.

To summarize, for each segment, we obtained a set of metrics, including GSR peaks per minute and number of peaks for measuring arousal, EEG metrics such as cognitive load to assess cognitive engagement, eye-tracking heatmaps for studying gaze patterns, and Affectiva-AFFDEX-derived emotions for understanding emotional responses. Importantly, the metrics were first calculated at the single-subject single-task level, and then averaged across subjects.

3.3 Results and Implications

3.3.1 Task 1 & 2: Compare the old and new dashboards. We compared eye-tracking and EEG data from Tasks 1 and 2 to assess differences in user engagement during visual exploration in both the old and new versions of the dashboard. Visual inspection of eye-tracking heatmaps and scan-paths revealed distinct attention patterns (see Fig. 2b & 2c), with the new dashboard showing a more organic attention pattern resembling natural reading flow. This organic attention pattern on the new dashboard may have contributed to reduced cognitive load, indicated by lower average EEG frontal theta PSD compared to the old one. The result suggests that clear

labels and explanatory text accompanying dashboard elements in the new version facilitated this natural reading flow. Moreover, the new dashboard's three separate tiles led us to examine EEG data variations related to each tile, highlighting how the tile with more written information increased cognitive load, suggesting that achieving a balance between visual and textual information is a crucial dashboard design consideration.

3.3.2 Task 3 to 6. In Task 3, participants created a new event on the dashboard. Eye-tracking data revealed an unexpectedly long average time-to-first-fixation (7.3 seconds) on the *Add Event* button, despite its direct visibility on the dashboard. This indicates that participants explored other areas before focusing on the button, suggesting a potential need for relocation. Regarding EEG data, we segmented it into three actions: (1) searching for the *Add Event* button, (2) adding the event (including selecting a time frame), and (3) observing changes. Participants took considerable time to locate the *Add Event* feature. The cognitive load increased when they attempted to add the event and further increased after the event was added. This indicates difficulties in both using the *Add Event* feature and identifying changes on the dashboard after event creation. Tasks 4 and 6 had notably longer times-to-first-fixation for locating the features to create a new view (Task 4), change and save dashboard settings (Task 5 & 6). Creating a new view involved sub-menu navigation, resulting in an average of 3 revisits to the menu and an average identification time of 22.8 seconds. Similarly, changing dashboard settings, also in a sub-menu, took an average of 27.8 seconds to locate. These findings indicate a potential mismatch between participants' expectations and the feature's placement.

4 CASE STUDY 2: ASSESSING A MOBILE APPLICATION IN THE AUTOMOTIVE SECTOR

4.1 Project Context

Our automotive industry client aimed to enhance their mobile application's homepage for better engagement and navigation. They designed four different homepage versions, and our task was to evaluate them for user engagement and cognitive load. Using biometric sensors, we aimed to provide insights for improving the user experience.

4.2 Methods

In this study, the objective was to conduct a comparative analysis of four versions of the application homepage, including the control version (i.e., the current homepage), Prototype A, B, and C. The client provided us with ProtoPie prototypes for each version. We used an iPhone Xs and three biometric sensors (EEG, GSR, and eye-tracking) similar to Case Study 1. The main difference was the mobile testing environment as compared to the desktop in Case Study 1. The phone was secured on a Tobii Mobile Testing Stand with a Tobii Pro Nano mounted above it. A screen recording via AV.io HD was used for eye-tracking analysis, providing insights into participants' gaze patterns and interactions with the interface.

4.2.1 Participants. Eight Participants took part in the study (4 female, mean age 47.1 ± 6.6). Although participants owned various



Figure 2: Case Study 1: (a) Experiment setup with the EEG, GSR and eye-tracking sensors; (b) The averaged eye-tracking heatmap on the old dashboard; (c) The averaged eye-tracking heatmap on the new dashboard

car brands, our main recruitment criterion was their prior experience with a car companion app, which they used daily or multiple times a week.

4.2.2 Experimental procedure. The **Step 1 to 3** of the experimental procedure closely resembled that of Case Study 1 (see Section 3.2.2), with the main difference being the use of a mobile device, which required some extra caution regarding eye-tracking calibration and participant positioning (see Fig. 3a). Once participants were prepared, they interacted with the prototypes and completing the tasks as **Step 4**. To mitigate potential learning effects, we systematically randomize the order of prototypes presented to each participant. After each prototype, participants filled in a survey to assess the ease of tasks. There were 5 minutes break before moving to the next prototype.

Participants completed four tasks for each homepage version. The tasks were carefully designed to assess participants' ease of use, cognitive load, and physiological arousal. These tasks are: (1) explore the homepage versions (to understand how participants engage with and visually navigate them); (2) locate the vehicle, access it, and start its engine (to evaluate how quickly and accurately participants locate and start a vehicle); (3) activate the air conditioning (to evaluate how easily participants use this functionality); (4) transition the vehicle to electric mode and initiate the charging process (to measure how easily participants use electric vehicle (EV) features).

4.2.3 Data preprocessing and analysis. For data preprocessing, analysis, and metrics calculation, we followed the same approach as in Case Study 1. We placed particular importance not only on the EEG-derived cognitive load metric but also on frontal alpha asymmetry (FAA) [28], which was calculated as the natural logarithm of the ratio between the PSDs of frontal electrodes F7 and F8 in the alpha frequency and served as a key indicator for task engagement. We also analyzed participants' subjective feedback on the designs and their self-assessments of performance in the surveys.

4.3 Results and Implications

We triangulated the data from EEG and GSR and observed that prototype A was associated with lower cognitive load and higher EEG FAA, while also leading to higher emotional intensity (i.e., overall higher GSR peaks per minute) compared to the other prototypes.

While prototype C also led to an overall high FAA, suggesting positive emotional valence, it was also associated with higher cognitive load compared to the other prototypes. In addition, we also delved into the individual tasks and use eye-tracking data to understand the causes behind certain trends indicated by the EEG and GSR data. We discuss the most significant results as follows.

EEG scan paths from **Task 1** suggested that the high cognitive load in Prototype C and the control version result from excessive textual information. This finding was supported by the survey responses, where three participants reported that the information on the homepage of these two version was not relevant. In **Task 2**, Prototypes A and B that include labels alongside icons, were associated with lower cognitive load compared to Prototype C and the control version. This suggests that icons with intuitive labels could reduce the cognitive burden on users, making it easier for them to interact with the interface. In **Task 3**, Prototype A resulted in the lowest cognitive load compared to the other prototypes. Since Prototype A was the only version that provides a visual cue inviting participants to long-press the button to complete the task, this explicitly visual cue potentially accounted for the lower mental workload observed. In **Task 4**, participants struggled to locate and change the vehicle in Prototype A because they overlooked the dropdown arrow positioned next to the vehicle's name. We observed an increase in participants' cognitive load measured by EEG.

5 CASE STUDY 3: USING BIOMETRIC SENSORS OUTSIDE THE LAB

5.1 Project Context

The market is witnessing a growing number of consumer-grade EEG and GSR devices that are portable and can be used wirelessly [26]. This presents an opportunity for conducting real-world biometric studies with increased ecological validity. This case study was a showcase conducted by two researchers (R1 and R2), aiming at demonstrating the capabilities of biometric sensors in revealing customers' in-store emotional journeys, attention patterns, and engagement with products or shop assistants for our client in the retail industry.

5.2 Methods

Researcher R1 played the role of a customer in the retail store, while R2 set up GSR sensors to measure emotional intensity of R1

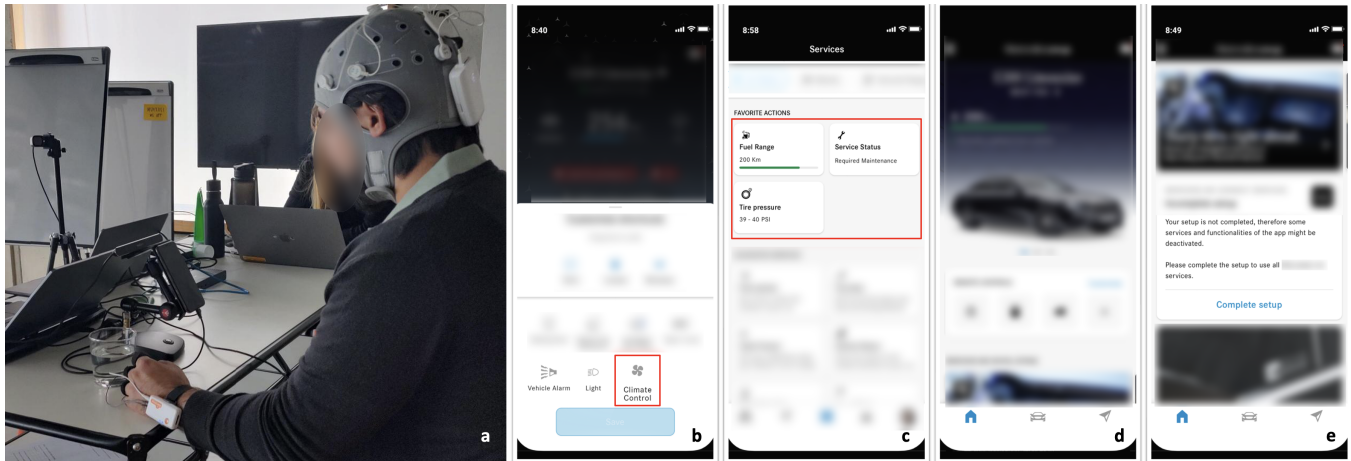


Figure 3: Case Study 2: (a) Experimental setup with the EEG, GSR, eye-tracking sensors, and a mobile stand. Design recommendations: (b) Offer visual indicators of affordances (Prototype A); (c) Use icons and organize elements to help users find what they are looking for (Prototypes A and B); (d) Incorporate a modern design approach (Prototype C); (e) Avoid including content on the homepage that is not directly related to users' goals (Prototypes C and the control version).

throughout the in-store journey. Additionally, R1 was equipped with eye-tracking glasses to gain insights into her visual exploration of the store space and interactions with displayed items and shop assistants. We broke down the experimental session into four different phases: (1) store layout exploration, (2) in-store product search, (3) product assessments, and (4) shop assistant consultation (see Fig. 1).

5.3 Results and Implications

Store layout analysis. Using eye-tracking heatmaps, we identified areas of attention and focus. The ease of access to store products, including layout and pathways, impacts customer engagement. For example, R1's attention went to products that were explicitly displayed along the pathway and products behind the pillars were ignored.

Store navigation. The GSR data indicated that the initial product that caught R1's attention did not generate emotional arousal due to crowded display arrangements and limited product information. R1 needed to bow down and move closer to the product multiple times. This observation indicates that engagement decreases when customers encounter obstacles in reaching products, highlighting the importance of accessible and obstacle-free product displays.

Product assessments. Through eye-tracking and GSR analysis, we gained insights into how specific products and their design elements captured R1's attention and emotional engagement. For example, while the foulard displayed on the central wall with back light is more visually pleasant, R1 mainly looked at and showed emotional engagement with the foulards hanging on the left rack, without fancy lighting. This observation invites further understanding of why visually pleasing displays do not win over more complex product assortments.

Shop assistant consultation. We monitored GSR activity to assess R1's reactions during a consultation. Frustration and increased

GSR activity were noted when the participant experienced delays and perceived a lack of attention from the saleswoman. This observation suggests neglecting customer service can result in frustration, decreased engagement, and diminished interest in products.

In summary, our showcase demonstrates to the client how biometric sensors can provide real-time insights into customers' emotional reactions and engagement, revolutionizing understanding of in-store user experiences.

6 DISCUSSION

6.1 Limitations and Opportunities

Due to client confidentiality, we cannot reveal the complete data and interfaces of their products. However, we have presented the advantages of small-sample biometric user research for supporting today's agile product development. Unlike scientific laboratory studies, which require a sufficient number of subjects for statistical power, in a commercial context, small-scale rapid testing with scientific rigor and quick feedback is more favored within the agile product development because it allows for refinement of designs based on immediate insights, accelerating the development process. We emphasize in-depth qualitative insights over quantitative measurements. These targeted insights (e.g., comparing two visual cues) often uncover nuanced issues that might be missed in larger studies. Our case studies demonstrate that we can leverage the insights gained from a limited pool of subjects to provide design recommendations.

6.2 Lessons Learned

6.2.1 Comparative Insights: Biometric versus Non-Biometric Methods. In our case studies, we used biometric methods to enrich our understanding of user experiences, revealing several differences from traditional non-biometric methods (e.g., questionnaires and observations). The primary benefit of biometric research is its ability to quantify user responses across a wide range of metrics. While

biometric methods offer objective data that traditional methods cannot easily capture, it is highly complex to generate real-time insights. In free-viewing scenarios, different users exhibit varied responses to stimuli, making it challenging to generalize findings, especially with small participant pools like those normally available in user experience research settings. It follows that the detailed insights are often limited to specific interactions and cannot be broadly applied easily. On the other side, traditional UX methods, while providing the context and subjective experiences that biometric methods lack, they are susceptible to personal biases and do not provide real-time data. To conclude, while proper rigorous comparisons of biometric and non-biometric methods are needed, we propose integrating biometric methods with traditional ones to gain a more complete and nuanced understanding of user experiences.

6.2.2 Define a baseline task for EEG data. One valuable lesson we learned from our studies is the importance of selecting an ideal baseline task to ensure the validity of our findings. To comprehend how design alterations affect EEG metrics like cognitive load, it is advisable to gather data related to a baseline task, preferably one that has a length similar to the tasks being assessed, which is straightforward and anticipated to be completed effortlessly. In our studies, we chose to use a simple EEG “resting state” baseline [7], which involves asking participants to sit quietly and comfortably and focus their gaze on a specific point on the screen. This strategy enables a comparative analysis between the baseline task and the cognitive tasks and facilitates a more precise evaluation of design-induced changes in cognitive load across various tasks. Having a resting-state baseline is particularly important in EEG research to account for differences in EEG baseline across participants.

6.2.3 Choose the appropriate sensors. Various user research scenarios call for specific physiological or behavioral parameters to be measured. For instance, EEG sensors are suitable for assessing cognitive load in tasks involving decision-making and information processing. GSR sensors are ideal for analyzing emotional responses, measuring stress, arousal, and sympathetic nervous system activity, making them valuable for emotional design and UX testing [3]. ECG sensors may be preferred over PPG sensors when precise heart rate measurements are necessary in healthcare, stress-related studies, or product development. PPG sensors are suitable for heart rate, blood volume, and blood oxygen saturation measurements, making them valuable for fitness trackers and health monitoring applications [34]. Eye-tracking metrics like time-to-first-fixation and revisits offer insights into user difficulties in searching information.

In addition to sensor functionality, participant comfort and acceptance are crucial for effective data collection. Opting for non-invasive or minimally invasive sensors that don't discomfort participants can enhance overall study quality. For instance, consumer-grade EEG devices often come as comfortable headsets or wearable headbands, eliminating the need for invasive procedures or medical supervision during setup [15]. Moreover, data synchronization and environmental suitability are critical factors when selecting sensors. Many of our studies integrate data from multiple sensors for a comprehensive understanding of user behavior and physiological responses through triangulation [30]. Effective synchronization is vital for coherent data and analysis, especially when using EEG, GSR, and eye-tracking sensors simultaneously. In field studies like

Case Study 3, portability and adaptability to noisy environments are essential.

6.2.4 Data segmentation for granular analysis. Segmenting the collected biometric data into shorter actions, rather than analyzing it as a whole, allowed us to conduct a more granular analysis. However, it is important to note that the segmentation can be challenging due to the variability in participants' actions. Therefore, we advise designing relatively simple tasks that do not become overly prolonged, enabling more consistent and meaningful insights from the analysis. For example, we can use eye-tracking data to segment a task into a sequence of brief actions, such as navigating to the product category page, locating the search bar, searching for Product A, and clicking on Product A for more information.

6.3 Standardization and Ethical Considerations in Biometric User Research

Zhu and Lv [35] have highlighted the current absence of a systematic guide, standard procedures, and benchmarking strategies in biometric user research, which poses challenges in maintaining consistency, reliability, and comparability of results. Standardization involves establishing consistent experimental protocols to ensure uniformity across studies and organizations. This process includes defining guidelines for data collection, preprocessing, metrics, and task design. Furthermore, it involves creating benchmarks for UX metrics, such as cognitive load, emotional valence, or usability, to assess the relationship between biometric data and user experience. Importantly, as UX research becomes an indispensable process for product development, companies conducting user studies should establish a research ethics committee to facilitate the ethical examination of biometric user research and other studies involving human subjects. To enable the widespread adoption of biometric methods, addressing various ethical challenges, such as privacy and equity, is crucial. Biometric data, such as EEG, GSR or eye-tracking, should be treated similarly to medical data, due to its sensitive and personal nature. It is fundamental to establish practices that ensure biometric data are collected, stored, and used in ways that respect user privacy. Moreover, future biometric research should aim for equitable representation across diverse user groups. An example is EEG research, as discussed by Choy et al. [6], which has shown systemic biases due to a lack of data from certain racial groups, leading to less generalizable findings and disparities.

7 CONCLUSION

We have presented three small-sample biometric user research studies suitable for time-constrained product development. These studies highlight how biometric methods complement traditional user research by providing real-time, objective insights into user reactions and emotions. Our key takeaways stress the importance of defining effective baseline tasks, choosing the right sensors, segmenting data based on simple activities, and emphasizing data triangulation for a comprehensive view. We also advocate for establishing research ethics committees and standardization in biometric user research for ethical, consistent, and reliable results across organizations.

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